

xCAT

Platform: Rocky Linux 8.10 — Master / Management Node **xCAT Version:** 2.17.0

Introduction to xCAT (Extreme Cluster Administration Toolkit)

1. What is xCAT?

xCAT is an open-source, scalable, distributed computing management software. Originally developed by IBM, it is now the industry standard for managing high-performance computing (HPC) clusters, clouds, and data centers.

Think of xCAT as the "**Orchestrator**" of your cluster. Instead of manually installing an OS on every single compute node with a USB drive, xCAT allows you to control, deploy, and monitor thousands of nodes from a single central point: the **Management Node (MN)**.

2. What Does xCAT Actually Do?

xCAT handles the entire lifecycle of a server, from the moment it is plugged into the rack until it is running parallel jobs. Its core functions include:

- **Provisioning:** Deploying operating systems (Rocky Linux, RHEL, Ubuntu, etc.) onto nodes via the network.
- **Hardware Management:** Remote power control (on/off/reset), firmware updates, and console access (SOL - Serial Over LAN) using protocols like IPMI or Redfish.
- **Inventory Management:** Maintaining a database of every node's MAC address, IP, hardware type, and rack location.
- **Hierarchy Support:** For massive clusters (10,000+ nodes), xCAT can use "Service Nodes" to offload work from the Management Node, ensuring the system never slows down.

3. Key Deployment Modes

One of the most important concepts in xCAT is how the OS is delivered to the nodes:

- **Stateful (Diskfull):** The OS is installed onto the node's local hard drive/SSD. Once installed, the node boots from its own disk.

- **Stateless (Diskless):** The OS lives in the RAM of the compute node. Every time the node reboots, it fetches a fresh "image" from the Management Node. This is highly popular in HPC because it ensures every node is identical and makes updates very easy.
- **Statelite:** A hybrid approach where the OS is in RAM, but specific persistent data (like logs or config files) is stored on an NFS server.

4. How the "Magic" Happens (The Workflow)

To provision a node, xCAT orchestrates several network services in a specific sequence:

1. **DHCP:** The node asks for an IP address. xCAT recognizes its MAC address and gives it a pre-defined IP.
2. **TFTP/HTTP:** The node downloads a small bootloader (like GRUB or iPXE).
3. **Kernel/Initrd:** The node downloads the Linux kernel and a temporary file system to start the installation.
4. **Post-Scripts:** After the OS is installed, xCAT runs scripts to configure things like SSH keys, NFS mounts, and Slurm automatically.

5. Prerequisites for Installation

Before installing xCAT, the environment must meet these specific conditions:

- **Dedicated Management Network:** You need a private network (e.g., 192.168.100.0/24) where the Management Node can talk to the nodes without interference from the public internet.
- **Static IP on MN:** The Management Node must have a static IP address.
- **Proper Name Resolution:** `/etc/hosts` or a DNS server must be configured so that every node name (e.g., `compute1`) resolves to an IP.
- **Repository Access:** You need access to the OS ISO/repositories and the xCAT online repositories to pull the latest packages.

- **Base OS:** A clean installation of a supported Enterprise Linux (like Rocky Linux 8.10).

Installation & Configuration

Step 1 — Set Hostname

Set the fully qualified hostname on the Master node before installing anything.

```
hostnamectl set-hostname hpc-master.hpc.local
```

Verify:

```
hostnamectl
```

Expected output:

```
Static hostname: hpc-master.hpc.local
```

Step 2 — Disable Firewall

xCAT requires open communication between the master and all nodes. The firewall must be fully disabled.

```
# Stop the firewall immediately
systemctl stop firewalld

# Disable it from starting on boot
systemctl disable firewalld

# Verify it is disabled
systemctl status firewalld
```

Expected output:

- firewalld.service
Loaded: loaded (/usr/lib/systemd/system/firewalld.service; disabled)
Active: inactive (dead)

Step 3 — Disable SELinux

SELinux can interfere with xCAT's TFTP, DHCP, and SSH operations. It must be disabled permanently.

3.1 — Disable SELinux immediately (without reboot):

```
setenforce 0
```

3.2 — Disable SELinux permanently:

```
# Edit the SELinux config file
sed -i 's/^SELINUX=.*SELINUX=disabled/' /etc/selinux/config
```

OR

```
vi /etc/selinux/config
```

Find The line - SELINUX=enforcing

Replace with

```
SELINUX=disabled
```

3.3 — Verify the change:

```
cat /etc/selinux/config
```

Expected output:

```
SELINUX=disabled
```

A full reboot is required for SELinux to be completely disabled.

Step 4 — Configure /etc/hosts

Add all cluster nodes to the hosts file on the Master node so name resolution works before DNS is configured.

```
vi /etc/hosts
```

Add the following lines:

```
# HPC Cluster Nodes
192.168.100.10    hpc-master    hpc-master.hpc.local
192.168.100.11    login         login.hpc.local
192.168.100.13    compute1     compute1.hpc.local
192.168.100.14    compute2     compute2.hpc.local
192.168.100.12    storage      storage.hpc.local
```

Verify:

```
ping -c 1 compute1.hpc.local
```

Step 5 — Update the System

Ensure the OS is fully up to date before installing xCAT.

```
dnf update -y
```

Reboot after update:

```
reboot
```

After reboot, verify SELinux is fully disabled:

```
getenforce
# Expected: Disabled
```

Step 6 — Add xCAT Repositories

xCAT requires two repositories: **xcat-core** (main packages) and **xcat-dep** (dependencies).

6.1 — Add xcat-core repository:

```
wget https://xcat.org/files/xcat/repos/yum/latest/xcat-core/xcat-core.repo \
-O /etc/yum.repos.d/xcat-core.repo
```

6.2 — Add xcat-dep repository for Rocky Linux 8 / RHEL 8:

```
wget https://xcat.org/files/xcat/repos/yum/xcat-dep/rh8/x86_64/xcat-
dep.repo \
-O /etc/yum.repos.d/xcat-dep.repo
```

6.3 — Verify repositories are added:

```
dnf repolist | grep -i xcat
```

Expected output:

```
xcat-core      xCAT 2.x for el8 x86_64
xcat-dep       xCAT 2.x for el8 x86_64 - Dependencies
```

Step 7 — Install xCAT

```
dnf install xCAT -y
```

This installs xCAT and all its dependencies including DHCP, TFTP, DNS, and other required services.

Verify installation:

```
# Load xCAT environment variables
source /etc/profile.d/xcat.sh

# Check xCAT version
lsxcatd -a
```

Expected output:

```
Version 2.17.0
This is a Management Node
dbengine=SQLite
```

Step 8 — Start and Enable xCAT Service

```
# Start xcatd
systemctl start xcatd

# Enable xcatd to start automatically on boot
systemctl enable xcatd

# Verify it is running
systemctl status xcatd --no-pager
```

Expected output:

- xcatd.service - xCAT management service
Active: active (running)

Step 9 — Configure xCAT Site Table

The site table holds global cluster-wide settings. These must be configured before provisioning any nodes.

```
# Set the Master node IP
chdef -t site master=192.168.100.10

# Set the cluster domain
chdef -t site domain=hpc.local

# Set nameserver (points to Master node)
chdef -t site nameservers=192.168.100.10

# Set upstream DNS forwarders (use your actual DNS or router IP)
chdef -t site forwarders=8.8.8.8,8.8.4.4

# Set the network interface xCAT will serve DHCP on
chdef -t site dhcpinterfaces="ens224"

# Set NTP server
chdef -t site ntpservers=192.168.100.10
```

Verify site table settings:

```
lsdef -t site
```

Step 10 — Import the Rocky Linux 8.10 ISO

Copy the Rocky Linux 8.10 DVD ISO to the Master node, then import it into xCAT.

```
# Import the ISO (xCAT will auto-detect and create OS images)
copycds /path/to/Rocky-8.10-x86_64-dvd1.iso
```

Verify OS images were created:

```
lsdef -t osimage
```

Expected output (similar to):

```
rocky8.10-x86_64-install-compute    (osimage)
rocky8.10-x86_64-install-service   (osimage)
rocky8.10-x86_64-netboot-compute    (osimage)
rocky8.10-x86_64-stateful-mgmtnode  (osimage)
rocky8.10-x86_64-statelite-compute  (osimage)
```

Step 11 — Define Nodes in xCAT

Add each node to xCAT's database with its MAC address, IP, group, and OS image.

11.1 — Add Compute Node 1:

```
mkdef -t node compute1 \
  groups=compute,all \
  mac=00:50:56:3A:18:A7 \
  ip=192.168.100.13 \
  netboot=xnba \
  arch=x86_64 \
  os=rocky8.10 \
  profile=compute \
  serialport=0 \
  serialspeed=115200
```

11.2 — Add Compute Node 2:

```
mkdef -t node compute2 \
  groups=compute,all \
  mac=00:50:56:2C:29:67 \
  ip=192.168.100.14 \
  netboot=xnba \
  arch=x86_64 \
  os=rocky8.10 \
  profile=compute \
  serialport=0 \
  serialspeed=115200
```

11.3 — Add Login Node:

```
mkdef -t node login \
  groups=login,all \
```



```
mac=00:50:56:34:93:8D \  
ip=192.168.100.11 \  
netboot=xnba \  
arch=x86_64 \  
os=rocky8.10 \  
profile=service \  
serialport=0 \  
serialspeed=115200
```

11.4 — Add Storage Node:

```
mkdef -t node storage \  
  groups=storage,all \  
  mac=00:50:56:26:46:3D \  
  ip=192.168.100.12 \  
  arch=x86_64
```

11.5 — Verify all nodes are defined:

```
lsdef -t node
```

Expected output:

```
computel    (node)  
compute2    (node)  
hpc-master  (node)  
login       (node)  
storage     (node)
```

Step 12 — Configure DNS and /etc/hosts

```
# Generate /etc/hosts entries for all nodes  
makehosts all  
  
# Generate and start DNS (named) configuration  
makedns -n  
  
# Verify named is running  
systemctl status named --no-pager
```

Step 13 — Configure DHCP

```
# Generate DHCP configuration for all defined nodes  
makedhcp -n  
  
# Verify dhcpd is running  
systemctl status dhcpd --no-pager  
  
# Confirm DHCP entries exist for nodes  
makedhcp -q all
```

Step 14 — Set Up SSH Key Trust

xCAT needs passwordless SSH from the Master node to all managed nodes.

```
# Generate and distribute SSH keys to all nodes
xdsh all --verify

# Test SSH connectivity
xdsh compute1,compute2,login whoami
```

Expected output:

```
compute1: root
compute2: root
login: root
```

Step 15 — Provision the Nodes (PXE Boot)

15.1 — Set the provisioning image for compute nodes:

```
nodeset compute osimage=rocky8.10-x86_64-install-compute
```

15.2 — Set the provisioning image for login node:

```
nodeset login osimage=rocky8.10-x86_64-install-service
```

15.3 — Power on / reboot nodes to begin PXE provisioning (check if it needs manual PXE Boot):

```
# If nodes support IPMI power control:
rpower compute1,compute2,login boot

# Monitor installation via console:
makegocons compute1
rcons compute1          # Press Ctrl+e, c, . to exit console
```

15.4 — Monitor provisioning status:

```
# Watch node status update in real time
watch -n 10 "tabdump nodelist | grep -E 'compute|login'"
```

Provisioning is complete when status changes to **booted**.

Step 16 — Verify Final Cluster Status

```
# Check all node statuses
tabdump nodelist

# Ping all nodes
ppping all

# SSH to all nodes
xdsh compute1,compute2,login whoami

# Check xCAT log for any errors
tail -50 /var/log/xcat/cluster.log
```